Electric Current Is Moving Charges

Electric Circuit Preamble

> SYNOPSIS
>
> You have learned that electrical charges generate electric fields that, in turn, influence (as attraction and repulsion force) other charges around them. Depending on the distance from the source charge, the affected charge feels the force (via an electric field, *E*) and, thus, has potential energy (by being in the potential, *V*). So far, we only talked about static charges, meaning stationary (not moving).

Charges want to move

Imagine charges sitting in a field (thus, potential). They want to move, if allowed! As we explore electric circuits (which make all electronic devices work, such as cell phones, computers, TVs, and so on), we need to look at moving charges! When charges are stationary, the energy is also stationary. For things to happen, we need a change in energy. The potential, measured in voltage (V), which is also known as the potential difference (note the word "difference"), tells us how much work the charges in the potential can do. This is, of course, related to the potential energy (*qV*, charge times potential). Just as an object higher up from the ground has larger potential energy than that on the ground (so it can do more damage when it falls), a large potential difference or the voltage indicates the charges that have high potential energy can do more work than that have a lower potential difference or lower voltage (so lower potential energy).

When charges are raised up higher in potential, meaning the charges are sitting in high potential (rocks are raised up high) and allowed to move (the rocks are released from the high position), the charges move toward the lower potential (the rocks fall). When the charges move from high voltage to lower voltage (the rocks fall from high to low), an electric current is created (falling rocks), which is measured in Amps (ampere, A = C/s), or the number of coulombs passing through in one second (if there was an unfortunate creature below the rocks, the number of rocks hitting the creature per second).

If the charges move down in potential (rocks fall) with minimum obstacles (freefalling rocks), charges can move at maximum speed, or the current is high (rocks falling at maximum speed). When the charges move at maximum speed, the maximum number of charges will pass through per second (maximum number of rocks per second hitting the creature). When the charges encounter restrictions (falling rocks hitting trees on the way down), the current decreases (the speed of rocks moving through decreases), and so the amount of charge passing through in one second is reduced (the number of rocks hitting the creature per second is reduced).

When obstacles restrict the charges' movements (we will discuss the nature of the obstacles soon), they move slower. This is exactly like the rocks experiencing friction (like air resistance or trees), which you know to be something that always impedes motion. When they move slower, the current is smaller, reducing the number of charges passing through. The obstacles create resistance, which is measured in Ohms (Ω, a Greek letter omega, and named after Ohm, who discovered Ohm's law, which we will delve into soon), and its effect is to slow down the current (the flow or movement of charges), in other words, reduce the current.

The picture of raising a rock (increasing potential or potential energy) and the falling rock (potential energy converting to other forms of energy, like movement energy, i.e., kinetic energy, or heat due to friction) is an excellent analogy for

charge in a field or potential that wants to move; it moves from a higher potential and drop toward lower potential like a rock rolling down a hill.

A more popular picture: water in a pipe

A more popular picture is water being pushed through pipes using a pump. Here, the pump creates pressure that can push the water higher, let's say, into a town water tower, which then flows down, moving through the pipes at lower heights. The pump is the battery, and the pressure (thus, height) in the tower is the voltage. So, the pressure is caused by the pump, moving water up and the water wanting to flow down; the potential difference is caused by the battery, moving charges up and the charges wanting to flow down. This is the voltage.

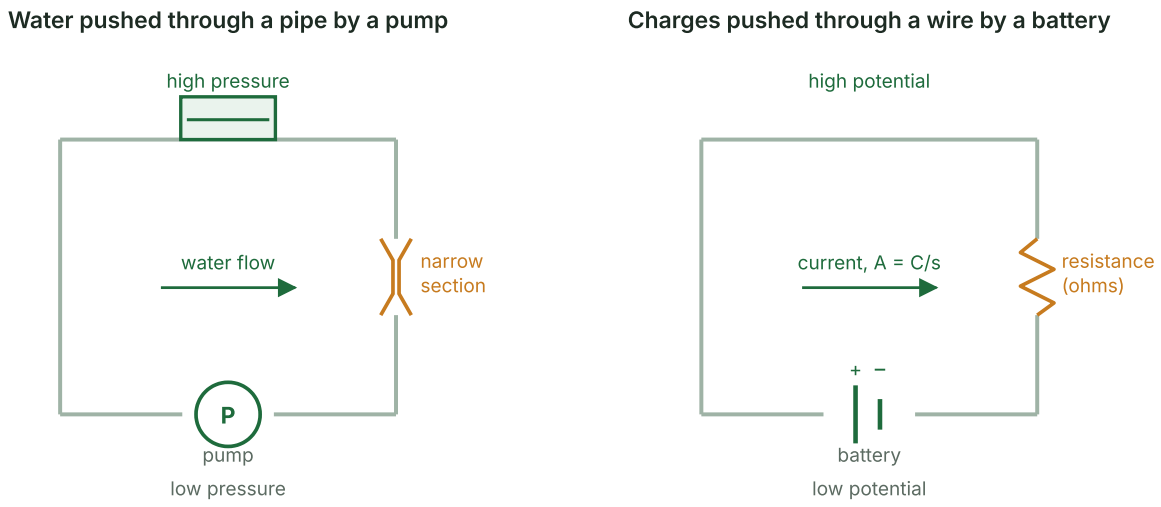


Figure 1. The same loop twice. The pump raises the water and the battery raises the charge; the height of the tower is the voltage; the pipe is the wire; the narrow section is the resistance. Each label is one row of the table below.

I summarize the concepts in an electric circuit using the water-pushed-through-the-pipe-by-a-pump analogy.

Water pushed through a pipe by a pump	Charges pushed through a wire by a battery
Water	Electrons
Pipe	Conducting wire
Pump (creates pressure)	Battery (creates potential difference)
Water flow (buckets per hour)	Electric current (A = C/s, coulombs per second)

Pressure (raises water height or millimeter of mercury, mmHg)	Voltage (V = J/C, joules per coulomb)
Water filter, narrow pipe sections	Resistance (ohms)

In this note, you will watch videos describing voltage, current, and resistance using the “water pushed through pipes using a pump” analogy. Answer questions based on what you learned.

PERFORMANCE EXPECTATIONS

1. Describe voltage, current, and resistance using the “water in pipe pushed by pump analogy.”

Instructions and Questions

1) Watch “What is electricity? — Electricity Explained (1)”

youtube.com/watch?v=ru032Mfsfig (Type “What is electricity? Electricity Explained” in the YouTube search bar).

1. What subatomic particles are responsible for electricity?

2. What forms an electric current?

3. How do you form a positively or negatively charged atom?

4. What are conductors? Give an example.

5. What are insulators? Give an example.

6. What is an electrical circuit?

7. What are the basic electrical elements and conditions for a continuous electrical circuit?

8. What is a short circuit?

9. Why is a short circuit dangerous?

10. How does a fuse prevent a short circuit?

2) Watch “What is ampere? — Electricity Explained (2)”

youtube.com/watch?v=xr68GxkiWFM (Type “What is ampere? Electricity Explained” in the YouTube search bar).

1. Why are metals excellent conductors?

2. What is an electric current?

3. How do you quantify current? What does it mean by a high current?

4. What is the unit of current?

5. What do you measure current with?

6. If you have 2 A of current, how many electrons pass through a point every second?

3) Watch “What is voltage? — Electricity Explained (3)”

youtube.com/watch?v=wmaobOy8r6I (Type “What is voltage? Electricity Explained” in the YouTube search bar).

1. How is the negative terminal in a battery akin to water in a water tower? Use the terms surplus/shortage and high or low pressure in your description.

2. How is the positive terminal in a battery akin to a syringe's plunger being pulled? Use the terms surplus/shortage and high or low pressure in your description.

3. Using the notion that nature always seeks the charge balance, why is there charge flow or current in a circuit?

4. Explain why you think we don't say just potential when we describe voltage but *potential difference*.

5. How does high voltage (potential difference) have a bigger capacity to do work? Use the water pressure and current analogy to explain your answer.

6. Why can't you light a regular household bulb with a 1.5 V battery? Explain your answer using the analogy of water pressure, flow, and heavy water mill.

7. The video, unfortunately, has a very grave error. What is it? Calculate the correct voltage using the same numbers for energy and charge.

8. What is a series connection? How does connecting three 1.5 V batteries in series affect the total voltage of the three batteries?



Figure 2. Three cells end to end, the + end of one to the – end of the next. **Bracket the three and write the voltage across all of them.**

Figures 2 and 3 together. Once you have marked the voltage on each, say in one line why the two answers are not the same.

9. What is a parallel connection? How does connecting three 1.5 V batteries in parallel affect the total voltage of the three batteries?

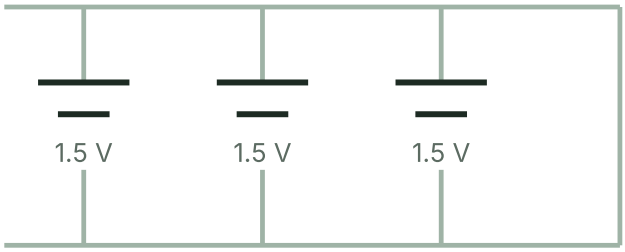


Figure 3. Three cells side by side between the same two rails. **Mark the voltage across the rails** — then say why it is not the answer you got for Figure 2.

Looking ahead

This note is the **preamble**: it names voltage, current and resistance and gives you one picture to hold them all together. What it does not do is put numbers on them. **Note 2** takes the current and the resistance into four simulations and makes the relation between them quantitative — Ohm's law. The "obstacles" in the paragraphs above get their real name there.